

Energy footprint of AI image as against a piece of traditional art.

Posted by u/Jackadullboy99 • 0 points • 12 comments

I'm starting to hear some concerns around the computational cost of increasing the "context window" (ie. short-term memory) of, for example, Chat GPT, and that it increases exponentially the more characters you want it to cover (5000 being the current limit for, I believe v3)..

It got me thinking of what the footprint might be for AI art, vs traditional or even digitally painted art. Let's assume the final product will end up as a physical canvas print/painting.. anyone care to hazard a guess as to how they might compare?

I guess you'd then have to then extrapolate to the scenario of billions of pieces being generated via AI. At least using the current models, might things start to look a little wasteful and ugly? Food for thought, maybe?

Doubtless the arms race for efficiency will fuel further innovations...

Comments

u/usrlibshare • 1 points • 2023-03-20 06:14 UTC

> Food fo thought maybe?

Yeah, no.

As long as we still live in car centric societies, where it's seen as normal to burn through 2 liters of gasoline to get half a liter of milk by driving an SUV to the corner store, ALOT has to change before I will start worrying about yhe CO2 footprint of generative AIs.

u/Sorry_Cattle1944 • 0 points • 2025-08-18 19:39 UTC

That is just so dumb, "I don't care about this bad thing because there's also another bigger bad thing."

like what, you can worry about both, because they're both bad

u/Evinceo • 1 points • 2023-03-19 16:03 UTC

Better use of all that GPU than the alternative, setting a giant pile of energy on fire to help people do crimes because central banks hurt your feelings.

u/unfamily_friendly • 4 points • 2023-03-19 08:16 UTC

What's more impactful? One minute of GPU working at 100% or 10 hours of GPU at 5%? Gamers beating both traditional and AI artists at this point

Can't speak for chatGPT tho. I heard language models are pretty power hungry

u/Vovann7b • 0 points • 2023-03-19 16:23 UTC

AI can't do what you want in one second. 10 hours at 100% GPU + training, how about that?

u/unfamily_friendly • 1 points • 2023-03-19 17:22 UTC

Idk who need 10 hours to make the pic they want. At my machine it takes 20 seconds per picture and i doubt i ever need more then 60 pics to get the desired results. It's like 20 minutes of high load, ignoring pauses and prompt editing

Training is more impactful but it's like, single 10 hours session divided by amount of people who will use it. 1 for 600 is 1 minute per person, which is even less.

I doubt someone makes new model every day, but i know a person who plays games 10 hours per day

u/WhatDoPlantsCrave • 10 points • 2023-03-19 05:49 UTC

TLDR I guessed total energy use per hour every day at 566,030 kWh for human artists and 392,425 kWh for AI generated images.

Human energy use at 14.25kWh per image and AI energy use at .10kWh per image.

AI image generator energy use per month 282,546,000kWh. That would power NYC for about 25 days.

Human-made about 40,000 images per hour, AI generated about 4,000,000 images per hour. 100 AI images generated for every human-made image.

This is not fact, it's a lot of wild guessing with some educated guesses thrown in. It would be fun to find out the real answers and how close or way off I was.

It's impossible to calculate without way more information, but sure we can guess. Let's try to average everything to a per hour basis and use kWh for energy. I'm not especially qualified to guess at this, but if anyone wants to modify my guesses please do so.

If it takes a human 10 hours to draw/paint on a tablet or computer, and it uses the average of .015kWh from googling, then that's .15kWh per complete image. That would probably be considered long for some artists, but not very long for others.

Many artists have lots of unfinished sketches that never get finished, so let's say 10 sketches per 1 completed illustration. Let's say these sketches get 20 minutes each so they cost .005kWh. For 10, we'll add it up to .05kWh.

So now we're at .2kWh per 1 complete image + 10 unrelated sketches.

Now we have to estimate how many of these images are completed by humans. How many illustrations are completed every hour across the world on average? Let's look at artstation, deviantart, and pixiv,

and instagram tagged #drawing and #painting and #digitalart to find how many are uploaded on average per hour, add them together.

So I roughly estimate about 6000 uploads to artstation in the last 24 hours = 250/hour. Rough estimate of 1600 on deviantart = 67/hour. 5000 on pixiv = 209/hour. Instagram tags I checked for the past hour instead of day because it's a nightmare, and it was #drawing 1000, #painting 1200, and #digitalart 1000 = 3200/hour. Obviously lots of these could be duplicates, reposts, AI generated, or unfinished sketches, and I didn't count them all by hand, so the numbers are just estimates. Add that all up and we get 3726 per hour. Let's round up to 4000 to err on the side of caution.

Most digital art never gets printed, just like AI generated images. But if we want to assume human artists print their art more often than AI generated images, then let's make this math even more guessey. At .08 kWh for an average print taking a minute on a desktop printer, and guessing that 1 in 1000 human illustrations get printed 5 times, that's 20 prints per hour, which is 1.6 kWh.

Let's account for a material cost for traditional media. It uses about 9376 kWh per ton of paper, average number of sheets per ton is 200,000 = .047 kWh per sheet. How about we triple that number to account for brushes, paints, pencils, and/or canvas = .141kWh. Let's guess 1/4 of our estimated images per hour number have the traditional media cost instead of the digital media cost, then that would be 1000 images x .141kWh = 141 kWh. And the 3000 remaining images x the digital art cost of .2kWh = 600kWh.

Let's also add in the human body energy use at 1.05 kWh x 13.3 hours per image = 13.965 kWh per image. For all 4000 images = 55,860 kWh.

141 (traditional) + 600 (digital) + 55,860 (human body) + 1.6 (printing) = 56,603kWh. Let's multiply this number by 10 to account for images that don't get posted on those websites = 566,030kWh energy used total for human-made 2D visual images made per hour.

Now we need to estimate compute power per AI generated image to compare. I'm estimating .1kWh per 512x512 image generated, based on estimates from google and reddit.

And now we can multiply that usage by an average number of images generated per hour from Midjourney, Dalle 2, and Stable Diffusion. Let's estimate 200 images per month per user of each, which is .28 images per hour. Midjourney 1 million users = 280,000 images per hour. Dalle 3 million users = 840,000 images per hour. Stable Diffusion impossible to calculate but google estimates 10 million users = 2,800,00 images per hour.

280,000 + 840,000 + 2,800,000 = 3,920,000 images per hour. Multiply by the .1kWh estimated energy per image = 392,000 kWh per hour used to generate AI images.

We can estimate maybe 1 in 100,000 are printed once? Which is 39.2 images printed x .08kWh = 3.136kWh. 392,000 kWh + 3.136 kWh = 392,003.136 kWh

Let's just take a wild guess that 1 in 10,000 are edited after generating for an additional 1 hour which adds 1.075kWh each of computer and human body energy for 392 images = 421.4kWh. 392,003.136 + 421.4 = 392,424.536kWh energy used total for AI images generated per hour.

If anyone read all of this, you probably used like .2625kWh of energy. I probably used at least 1.05kWh of energy thinking about this.

Edited to add the 4th line.

u/diditforthevideocard • 1 points • 2024-07-03 16:32 UTC

Thanks for this. Out of curiosity where do you get your figures--for example how many MidJourney images are created per day?

u/wholemonkey0591 • 1 points • 2024-02-22 16:06 UTC

What about energy used to make and support NFT's.

u/Evinceo • 3 points • 2023-03-19 15:34 UTC

/r/theydidthemath

u/Tiamatium • 1 points • 2023-03-19 04:03 UTC

Oh wow, that argument is just so... Stupid. No offence, but few seconds of GPU time (that's what's needed) would use very little energy, especially if we compare it to digital art, which would require hours of GPU time. Literally same goes for ChatGPT generated excel sheet Vs human one, it just does it in seconds

u/Jackadullboy99 • 1 points • 2023-03-19 05:53 UTC

Well, you tell me. Is that all the work that's being done? All local processing? It was presented as a question...perhaps a "stupid" one that no sensible person would have any business posing?

u/Evinceo • 3 points • 2023-03-19 15:59 UTC

Training the models is very compute intensive (billions of dollars in compute) but only needs to be done a tiny number of times (once the model is trained millions of people can use it on their home PCs.) The marginal cost of each imagine is small. I am worried about a future where people are scratch training a ton of huge models all the time, but as of now that's not the case.

u/wswordsmen • 1 points • 2023-03-19 03:27 UTC

Depends on a lot. When I run SD my computer seems to draw around 600W of power while generating an image. I normally do about 540x960, half of 1080x1920 and it takes about 15s per image. Normal use of the computer is closer to 100W. So overall it probably is much better OK a per image basis, however it probably takes a lot more images to produce good results so the fact no one could produce work in 90s isn't a fair comparison.

u/zfreakazoidz • 3 points • 2023-03-19 02:17 UTC

I mean all tech tends to leave big footprints. As much as we all tend to enjoy futuristic scenarios where our cyberpunk cities are amazing. It does all come at a cost. Unless of course we eventually get fusion power plants created.